Frequency Distribution Charts

- Histogram
- Frequency Polygon
- Smoothed frequency curve
- Ogives or Cumulative frequency curves.

Histogram

- Set of vertical bars whose areas are proportional to the frequencies represented. Vertical bars have class interval distance as its width.
- Variable is taken on x-axis and frequencies are taken on y-axis.
- Histogram can not be made with open end classes.
- When class intervals are unequal, corrections need to be made ie. finding relative frequency density for each class (frequency/width of the class). A class having lowest class interval is taken and frequencies are adjusted. For eg if one class interval is twice the width of lowest class interval, its frequency is to be divided by two.
- When only midpoints of class intervals are given then upper and lower limits need to be ascertained.

Frequency Polygon

- Two ways of constructing frequency polygon:
  - Draw a histogram and join by straight lines the mid points of upper horizontal side of each rectangle. Polygons are closed on both ends by extending the lines to the base line.
  - Take the midpoints of various class intervals and plot the frequency corresponding to each point and join them.
  - Adjustments same as in histograms need to be made in frequency polygon when there are unequal class intervals.
  - Frequency polygons can not be used for distributions having open end classes.
- Advantages of frequency polygon over histogram:
  - Frequency polygons of several distributions can be plotted and compared on the same axis.
  - Frequency polygon is simpler.
  - It sketches an outline of data pattern more clearly.
  - Polygon becomes increasingly smooth as number of observations increases.

Smoothed Frequency Curve

- Steps:
  - Draw a histogram.
  - Draw a polygon by joining mid points of upper horizontal sides of rectangles.
  - Smooth the polygon.
- Goal is to eliminate accidental variations that might be present.
- Curve should look as regular as possible and sudden turns should be avoided.

Cumulative Frequency Curve (Ogive)

- It is a curve obtained by plotting cumulative frequencies (sum of frequencies).
- Two methods of constructing ogives:
  - 'Less than' method: We start with upper limits of classes and go on adding the frequencies. As a result we get a rising curve.
  - 'More than' method: We start with lower limits of the classes and from the frequencies we subtract the frequency of each class. As a result we get a declining curve.
- In this method it is very easy to answer questions like: "how many students obtained marks less than a particular value".

- Utility of Ogives:
  - To determine and portray the number of cases above or below a given value.
  - To compare two or more frequency distributions.
  - To determine certain values graphically such as median, quartiles etc.
  - Ogives are not very simple to interpret.
- Limitations of graphs/charts:
  - They can present only approximate values.
  - They can approximately represent only limited amount of information.
  - They are intended for general public. They are not of much help for statisticians for analysing data.
  - They can easily be misinterpreted.

Reference(s)

- Gupta, S. P., and M. P. Gupta. *Business statistics*. Sultan Chand & Sons, 1983.